Table 3. Salivary Gland Tumor Subtypes, Molecular Drivers, and Organoid-Relevant Targets

Tumor subtype	Histology	Molecular features	Relevant pathways	Candidate therapy /target	PDO evidence / limitation	Refs
Mucoepidermoid carcinoma (MEC)	Squamoid, mucous, intermediate cells	CRTC1/3–MAML2 fusion	CREB-associated transcription; autocrine AREG–EGFR axis	EGFR-directed agents (proposed)	MEC PDOs reported; EGFR-targeted screening remains a proposed use, not direct PDO evidence.	7,17,101,104–106,113
Adenoid cystic carcinoma (ACC)	Cribriform, tubular, solid	MYB–NFIB / MYBL1–NFIB; chromatin and NOTCH alterations	MYB/MYBL1 programs; NOTCH in selected tumors	NOTCH/pathway-directed agents (selected cases; exploratory)	ACC PDOs reported; verify fusion/genotype retention and passage-associated changes.	7,16,17,101,107–110,112,113
Secretory carcinoma	Microcystic, tubular	ETV6–NTRK3 fusion	Constitutive TRK signaling	TRK inhibitors (e.g., larotrectinib, entrectinib)	Direct TRK-inhibitor sensitivity demonstrated in a secretory carcinoma PDO.	7,102
Salivary duct carcinoma (SDC)	Apocrine, high-grade	AR expression; HER2 amplification; PI3K–AKT–mTOR alterations	AR, HER2 and PI3K–mTOR signaling	AR antagonists; HER2-directed and pathway inhibitors	Genotype-oriented SDC PDO testing reported; reassess driver/target retention, including HER2.	7,16,17,101,113,128
Epithelial–myoepithelial carcinoma (EMC)	Biphasic epithelial–myoepithelial	HRAS mutation (subset)	RAS–MAPK signaling	RAS/MAPK pathway targeting (investigational)	Subtype-specific PDO and drug-response evidence remains limited.	7,17,111
CaExPA; polymorphous adenocarcinoma	Variable / rare	Heterogeneous; subtype-dependent	Subtype-dependent	No PDO-validated target specified	Systematic subtype-specific PDO evidence remains limited.	7,100

PDO, patient-derived organoid; DDR, DNA damage response; AR, androgen receptor; RTK, receptor tyrosine kinase; CaExPA, carcinoma ex pleomorphic adenoma.